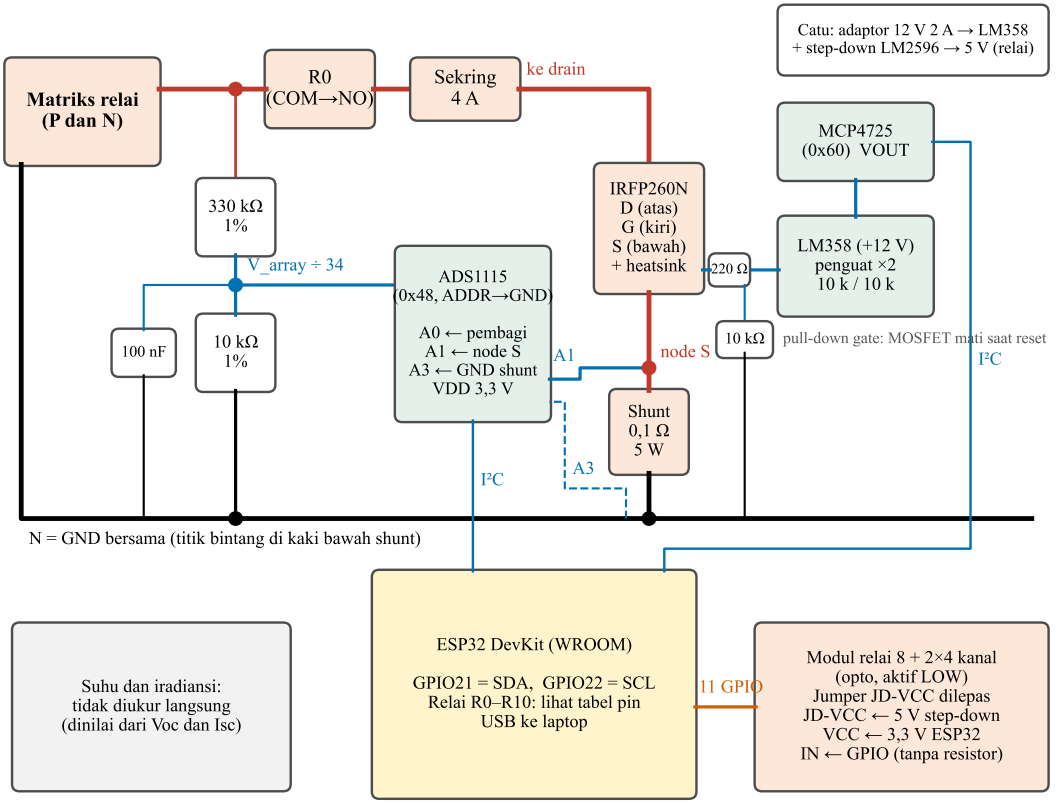
Catu: adaptor 12 V 2 A → LM358
+ step-down LM2596 → 5 V (relai)
Matriks relai
(P dan N)
R0
(COM→NO)
Sekring
4 A
ke drain
MCP4725
(0x60) VOUT
IRFP260N
D (atas)
G (kiri)
S (bawah)
+ heatsink
330 kΩ
1%
V_array ÷ 34
ADS1115
(0x48, ADDR→GND)
A0 ← pembagi
A1 ← node S
A3 ← GND shunt
VDD 3,3 V
220 Ω
LM358 (+12 V)
penguat ×2
10 k / 10 k
100 nF
10 kΩ
1%
A1
node S
10 kΩ
pull-down gate: MOSFET mati saat reset
I²C
Shunt
0,1 Ω
5 W
A3
I²C
N = GND bersama (titik bintang di kaki bawah shunt)
Suhu dan iradiansi:
tidak diukur langsung
(dinilai dari Voc dan Isc)
ESP32 DevKit (WROOM)
GPIO21 = SDA, GPIO22 = SCL
Relai R0–R10: lihat tabel pin
USB ke laptop
11 GPIO
Modul relai 8 + 2×4 kanal
(opto, aktif LOW)
Jumper JD-VCC dilepas
JD-VCC ← 5 V step-down
VCC ← 3,3 V ESP32
IN ← GPIO (tanpa resistor)